

Martin He

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EDUCATION

Carnegie Mellon University

Pittsburgh, PA

Master of Computational Design (Machine Learning + CS focus)

Expected 2027

Bachelor of Architecture

Dec 2020

- **Relevant CS Coursework:** Imperative Computation (C), Fundamental of Programming(Python), Probability Theory (S26), Intro to Deep Learning (S26), Math for Machine Learning, Intro to Computer System(progress)

EXPERIENCE

EOA Architects, PLLC

Nashville, TN

Computational Lead

Feb 2022 – Dec 2024

- Developed and deployed Python/Revit API tools automating repetitive architectural production and design workflows.
- Built parametric tools for generating, evaluating, and comparing design options across active projects.
- Led computational development across multidisciplinary teams, translating architectural workflows into reusable software.

PROJECTS

Forecasting Models for Rare High-Impact Events in Electricity Markets

2026

- Developed an end-to-end time-series forecasting pipeline in Python/PyTorch for hourly ERCOT electricity prices, benchmarking ARIMA against LSTM architectures for rare price-spike prediction.
- Designed cost-sensitive learning strategies using weighted loss, continuous weighting, and oversampling for rare, high-impact events.
- Demonstrated that a weighted LSTM outperformed vanilla LSTM and ARIMA by balancing spike detection and false positives.

Integrated AEC Design & Real Estate Feasibility Platform

2026 - Ongoing

- Python · RhinoCommon · REST APIs · React · Financial Modeling
- Developing a feasibility platform connecting 3D architectural massing with area, program, cost, and financial analysis.
- Engineered geometry pipelines for floor slicing, Boolean operations, spatial classification, efficiency calculations, and fallback handling.
- Architected modular pipelines integrating geometry, project timelines, monthly cash flows, development costs, and investment metrics.
- Building an interactive frontend for comparing design options and their impact on financial feasibility.

TEACHING

Praxis 1 Design Studio at Carnegie Mellon University

Pittsburgh, PA

Guest Juror for Praxis 1 Design Studio

Aug 2025 – Dec 2025

SKILLS & CERTIFICATE

Software Languages: Python, C

Libraries and Tools: NumPy, Linux, Autodesk Revit API, Pytorch, Panda

Concepts: Data Structures & Algorithms, Machine Learning, Optimization, Data Analysis